المحاضرة الأولى Type of errors

Errors

Occurrence of error is unavoidable in the field of scientific computing. Instead, numerical analysts try to investigate the possible and best ways to minimise the error. The study of the error and how to estimate and minimize it are the fundamental issues in *error analysis*.

Error Analysis

In numerical analysis we approximate the exact solution of the problem by using numerical method and consequently an error is committed. The numerical error is the difference between the exact solution and the approximate solution.

Definition (Numerical Error): Let x be the exact solution of the underlying problem and x^* its approximate solution, then the error (denoted by e) in solving this problem is

$$e = x - x^*$$

Absolute and Relative Errors

Definition (Absolute Error): The absolute error $\hat{e}$ of the error e is defined as the absolute value of the error e

$$\hat{e} = |x - x^*|.$$

Definition (Relative Error): The relative error $\tilde{e}$ of the error e is defined as the ratio between the absolute error $\hat{e}$ and the absolute value of the exact solution x

$$\tilde{e} = \frac{\hat{e}}{|x|} = \frac{|x - x^*|}{|x|}, \quad x \neq 0.$$

Example: Let $x = 3.141592653589793$ is the value of the constant ratio π correct to 15 decimal places and $x^* = 3.14159265$ be an approximation of x.

Compute the following quantities:

a. The error.

b. The absolute error.

c. The relative error.

Solution:

a. The error

$$e = x - x^* = 3.141592653589793 - 3.14159265 = 3.589792907376932e - 09$$
$$= 3.589792907376932 \times 10^{-9} = 0.000000003589792907376932.$$

b. The absolute error

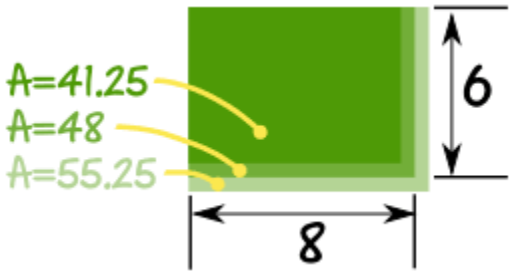
$$\hat{e} = |x - x^*| = |3.141592653589793 - 3.14159265| = 3.589792907376932e - 09.$$

c. The relative error

$$\tilde{e} = \frac{\hat{e}}{|x|} = \frac{|x - x^*|}{|x|} = \frac{3.141592653589793 - 3.14159265}{3.141592653589793}$$
$$= \frac{3.589792907376932e - 09}{3.141592653589793} = 1.142666571770530e - 09.$$

Example: Alex measured the field to the nearest meter, and got a width of 6 m and a length of 8 m.

Measuring to the nearest meter means the true value could be up to **half a meter** smaller or larger.



The width (w) could be from 5.5m to 6.5m: $5.5 \leq w < 6.5$

The length (l) could be from 7.5m to 8.5m: $7.5 \leq l < 8.5$

The area is width × length: $A = w \times l$

The smallest possible area is: $5.5m \times 7.5m = \mathbf{41.25\ m2}$

The measured area is: $6m \times 8m = \mathbf{48\ m2}$

And the largest possible area is: $6.5m \times 8.5m = \mathbf{55.25\ m2}$

$$41.25 \leq A < 55.25$$

Absolute, Relative and Percentage Error

The only tricky thing here is ... **which** is the absolute error?

- From 41.25 to 48 = **6.75**
- From 48 to 55.25 = **7.25**

Answer: pick the biggest one! So:

$$Absolute\ Error\ =\ 7.25\ m2$$

$$Relative\ Error = \frac{7.25m^2}{48m^2} =\ 0.151\ldots$$

$$Percentage\ Error\ =\ 15.1\%$$

Roundoff and Truncation Errors

Computers represent numbers in finite number of digits and hence some quantities cannot be represented exactly. The error caused by replacing a number a by its closest machine number is called the **roundoff error** and the process is called **correct rounding**.

Truncation errors also sometimes called **chopping errors** are occurred when chopping an infinite number and replaced it by a finite number or by truncated a series after finite number of terms.

Example: Approximate the following decimal numbers to three digits by using rounding and chopping (truncation) rules:

1. $x_1 = 1.34579.$
2. $x_2 = 1.34679.$
3. $x_3 = 1.34479.$
4. $x_4 = 3.34379.$
5. $x_5 = 2.34579.$

Solution:

i. Rounding:

a) $x_1 = 1.35.$

b) $x_2 = 1.35.$

c) $x_3 = 1.34.$

d) $x_4 = 3.34.$

e) $x_5 = 2.35.$

ii. Chopping:

a) $x_1 = 1.34.$

b) $x_2 = 1.34.$

c) $x_3 = 1.34.$

d) $x_4 = 3.34.$

e) $x_5 = 2.34.$

Example: π =3.141592654

مثلا نحسب خطأ التدوير لل π عند اربع مراتب π =3.1416

ونحسب خطأ القطع او البتر لل π عند اربع مراتب π =3.1415

Example:

x	Round-by-chop	Roundoff Error	Round-to-nearest	Roundoff Error
1.649	1.6	0.049	1.6	0.049
1.650	1.6	0.050	1.6	0.050
1.651	1.6	0.051	1.7	-0.049
1.699	1.6	0.099	1.7	-0.001
1.749	1.7	0.049	1.7	0.049
1.750	1.7	0.050	1.8	-0.050